22CSE19

BIG DATA ANALYTICS
(Professional Elective-I)

Instruction	3 Hours per week
Duration of SEE	3 Hours
SEE	60 Marks
CIE	40 Marks
Credits	3

PREREQUISITE: Proficiency in python programming, Database fundamentals

COURSE OBJECTIVES: This course aims to
1. Introduce the importance of Big Data and the role of the Hadoop framework in analyzing large datasets.
2. Gain practical knowledge in working with Hadoop ecosystem tools such as MapReduce,pig, hive, Spark.
3. Provide hands-on experience in using Spark for real-time stream processing.

COURSE OUTCOMES: After the completion of this course, the student will be able to
1. Define and describe the types, characteristics, and challenges of Big Data and the need for its analysis.
2. Analyze the differences between traditional systems and Hadoop and explain the functions of core components like HDFS and YARN.
3. Develop MapReduce programs and analyze different join strategies and pipeline models for big data processing.
4. Design and execute queries using Hive and Pig to extract, transform, and analyze large datasets.
5. Implement scalable data processing applications using Apache Spark and evaluate its performance over traditional frameworks

CO-PO Articulation Matrix

PO/PSO CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO 1	PSO 2	PSO 3
CO 1	2	3	1	3	3	3	-	3	3	2	3	2	2	-
CO 2	3	3	3	3	3	2	-	3	3	2	3	2	2	2
CO 3	3	3	3	3	3	2	-	3	3	2	3	2	2	2
CO 4	3	3	3	3	3	2	-	3	3	2	3	1	2	3
CO 5	3	3	3	3	3	3	-	3	3	2	3	2	3	3

1 - Slightly, 2 - Moderately, 3 – Substantially

UNIT-I

Introduction to Big Data: Data and its types: Unstructured, Semi-structured, Structured – Sources of data – Evolution and Definition of Big Data – Characteristics(3Vs/5Vs) and Challenges – Need for Big Data, , Big data integration process, Applications.Overview of Business Intelligence, Data Science, Analytics – Typical analytical architecture – Types of Analytics (Descriptive, Predictive, Prescriptive)

UNIT-II

Hadoop Ecosystem: Requirement of Hadoop Framework - Design principle of Hadoop –Comparison with other system (SQL, RDBMS) - Core ccomponents of Hadoop – Architecture -Hadoop 1 vs Hadoop 2 – HDFS operations, Data ingestion layer, ETL and ELT, Ingestion tools in Hadoop ecosystem, Data ingestion types.

UNIT-III

Big Data Frame Works: Hadoop Framework: Introduction to MapReduce, Processing data with Hadoop using MapReduce, Map Reduce Programming: I/O formats, Map side join-Reduce Side Join,

Secondary Sorting-Pipelining MapReduce jobs. Introduction to YARN, Architecture, Managing Resources and Applications with Hadoop YARN.

UNIT-IV

Hive and Pig: Introduction to Hive: What is Hive, Hive Architecture, Hive data types, Hive file formats, Hive Query Language (HQL), RC File implementation, User Defined Function (UDF).
Introduction to Pig: What is Pig, Anatomy of Pig, Pig on Hadoop, Data types in Pig, Running Pig, Execution Modes of Pig, HDFS Commands, Relational Operators, Eval Function, Complex Data Types, Piggy Bank, User Defined Function, Pig Vs Hive.

UNIT-V

Apache Spark Framework: Introduction to Apache spark-Design principles, Advantages and Disadvantages, Layered architecture, Programming with RDDs: Create RDD- Spark operations - transformations, actions-DAG in Apache spark.
Case Study: Analyze student academic data to find trends, subject difficulty levels, and top performers using big data tools.

TEXT BOOKS:

1. S. Chandramouli, 'Big Data Analytics', University press,2024 edition.
2. Seema Acharya, Subhashini Chellappan, "Big Data and Analytics", Wiley India Pvt. Ltd., 2016.

SUGGESTED READING:

1. Tom White, 'Hadoop: The Definitive Guide', 4th Edition, O'Reilly Media Inc, 2015.
2. Thilinagunarathne, 'Hadoop MapReduce v2 Cookbook', 2nd Edition, Packet Publishing, 2015.
3. Chuck Lam, Mark Davis, Ajit Gaddam, 'Hadoop in Action', Manning Publications, 2016.
4. Alan Gates, 'Programming Pig', O'Reilly Media Inc, 2011.

ONLINE RESOURCES:

1. http://www.planetcassandra.org/what-is-nosql
2. http://www.iitr.ac.in/media/facspace/patelfec/16Bit/index.html
3. https://class.coursera.org/datasci-001/lecture
4. http://bigdatauniversity.com